

A Cryo-CMOS Triple Tail Comparator with Capacitive Over-Neutralization to Suppress Freeze-Out Induced Hysteresis

Bram Veraverbeke , Graduate Student Member, IEEE, Filip Tavernier , Senior Member, IEEE

Abstract—Dopant freeze-out severely increases the bulk resistance of cryogenic bulk CMOS transistors by up to $10^6\times$ at 4.2 K compared to room temperature. This work describes, for the first time in the literature, how this increased bulk resistance introduces a memory effect in the latch of dynamic comparators, which leads to hysteresis. To measure this hysteresis reliably in the presence of noise, a statistical characterization procedure is developed. For a 40 nm bulk CMOS strongARM comparator with an input-referred noise voltage of $348\mu\text{V}_{\text{RMS}}$, a hysteresis voltage $>898\mu\text{V}$ is measured at 6 K, substantially deteriorating the precision. Therefore, this paper introduces a triple tail comparator with capacitive over-neutralization to increase the preamplification gain, suppressing the hysteresis $>6\times$ to only $141\mu\text{V}$.

Index Terms—Cryo-CMOS, dopant freeze-out, bulk resistance, memory effect, hysteresis, strongARM, dynamic comparator

I. INTRODUCTION

CRYO-CMOS has recently gained much attention thanks to emerging applications, such as quantum computing, particle physics experiments, and gravitational wave detectors [1], [2], [3], [4], [5], [6], [7]. To date, cryogenic characterization and modeling of modern CMOS processes primarily focus on DC-characteristics [6], [7], [8]. However, these are insufficient for analog-to-digital converters (ADCs), which contain many switching circuits such as dynamic comparators, since any inaccuracy due to transient effects in the comparator affects the full ADC performance. Still, in the literature, the characterization of cryogenic comparators is limited to a noise analysis of a strongARM comparator for a 7-bit ADC [9].

This paper shows, for the first time in the literature, that cryogenic bulk CMOS comparators such as the prevalent strongARM comparator in Fig. 1 suffer from hysteresis in their latch. This hysteresis is induced by a severe increase ($10^6\times$) of the bulk resistance due to dopant freeze-out [6], [10], [11]. The presented work provides additional insights into cryogenic bulk CMOS behavior and advances the state-of-the-art in three ways. Firstly, it describes in detail how dopant freeze-out introduces hysteresis in cryogenic comparators. This article focuses on dynamic comparators, but the same concepts can be applied to assess the impact of dopant freeze-out on any cryogenic circuit with fast switching transients.

Bram Veraverbeke and Filip Tavernier are with the Department of Electrical Engineering, KU Leuven, 3001 Leuven, Belgium. (e-mail: bram.veraverbeke@kuleuven.be, filip.tavernier@kuleuven.be). This work was supported by the Research Foundation-Flanders (FWO) under grant 11N0523N.

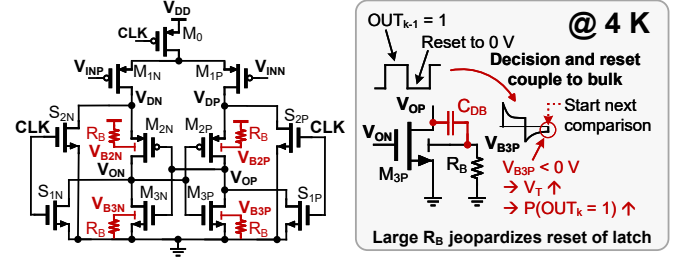


Fig. 1. The conventional strongARM (SA) comparator. The increased bulk resistance R_B jeopardizes a proper reset of the latch.

Secondly, a statistical characterization procedure is developed to reliably measure the hysteresis voltage in the presence of noise. Finally, a triple tail comparator with capacitive over-neutralization (TTCO) is proposed to suppress the hysteresis $>6\times$ by boosting the preamplification gain.

The remainder of this brief is organized as follows. Section II discusses the dopant freeze-out introduced hysteresis. Next, Section III introduces the proposed TTCO comparator. Sections IV and V discuss the custom cryogenic measurement setup and the statistical characterization procedure, respectively, after which the measurement results are presented in Section VI. Finally, a conclusion is formulated in Section VII.

II. DOPANT FREEZE-OUT INDUCED HYSTERESIS

The strongARM (SA) comparator [9], [12] in Fig. 1 is a popular topology thanks to its fast, energy-efficient, and robust operation. It comprises a capacitive preamplifier ($M_{0,1}$) and a latch ($M_{2,3}$). During the reset phase ($\text{CLK} = V_{\text{DD}}$), the preamplifier and latch outputs are discharged to V_{SS} through the reset switches ($S_{1,2}$). In the comparison phase ($\text{CLK} = V_{\text{SS}}$), the tail transistor M_0 is turned on, gradually charging V_{DP} and V_{DN} to V_{DD} at a slightly different rate, which generates some gain before the latch turns on. After the comparison phase, the next reset phase should erase all information from the previous decision. At room temperature, a proper reset is easily guaranteed by correctly sizing $S_{1,2}$.

At cryogenic temperatures, however, dopant freeze-out severely increases the substrate resistance by up to $10^6\times$ at 4.2 K compared to room temperature [6], [10], [11]. This leads to a large and unpredictable bulk resistance R_B , as highlighted in Fig. 1. Such a high resistance leaves the bulk vulnerable for capacitive coupling of the latch outputs through the drain-bulk capacitance C_{DB} . The bulk voltage, and consequently

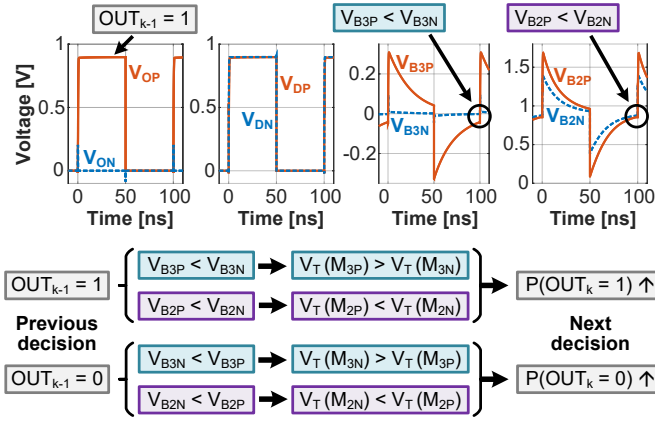


Fig. 2. Transient simulations of the preamplifier and latch outputs, and the bulk voltages of the latch transistors in the conventional SA comparator with $R_B = 15 \text{ M}\Omega$ and $f_{\text{clk}} = 10 \text{ MHz}$ (top). V_T -modulation through the bulk results in hysteresis that favors the same decision as the previous one (bottom).

the threshold voltage V_T , of the latch transistors $M_{2,3}$ is thus modulated by the previous comparator decision. In this way, information from the previous comparison leaks to the next comparison phase, jeopardizing a proper reset of the latch, which leads to hysteresis.

Since foundry models do not support cryogenic temperatures, the effects of dopant freeze-out are emulated by placing an additional large R_B between the bulk terminal and V_{DD} or V_{SS} . The transient simulations with room temperature models and an R_B of $15 \text{ M}\Omega$ in Fig. 2 show that the bulk voltages of $M_{2,3}$ are heavily disturbed by the large voltage swing at $V_{OP/N}$ and $V_{DP/N}$. Even at a low clock frequency f_{clk} of only 10 MHz , the bulk voltages cannot recover to their DC-values by the end of the reset phase. When the previous comparator decision was positive ($OUT_{k-1} = 1$), V_{B3P} will be lower than V_{B3N} at the start of the next comparison, while V_{B2P} will be lower than V_{B2N} . Hence, M_{3P} will have a higher V_T than M_{3N} , whereas M_{2P} will have a lower V_T than M_{2N} . This increases the probability of a positive decision during the next comparison. Due to the symmetrical topology of the comparator, a previous negative decision ($OUT_{k-1} = 0$) will analogously increase the probability of a negative decision during the next comparison. Consequently, the increased bulk resistance at cryogenic temperatures could introduce a noticeable hysteresis in the latch. The swing at $V_{DP/N}$ will also disturb the bulk of the input pair, but since both V_{DP} and V_{DN} have the same swing from V_{DD} to V_{SS} , this is mainly a common-mode disturbance. Hence, it does not significantly affect the next decision.

As the freeze-out induced hysteresis favors the same decision as the previous one, it can be modeled by an input-referred voltage source V_{HYS} whose sign depends on the previous decision, as shown in Fig. 3. Simulations show that V_{HYS} may rise substantially above the noise level of $350 \mu\text{V}_{\text{RMS}}$ for bulk resistances above $10 \text{ M}\Omega$ or clock frequencies above 10 MHz , so that the precision of the SA comparator is bounded by V_{HYS} instead of noise. Such a large hysteresis can introduce significant errors in an ADC. For example, in a successive-approximation-register (SAR) ADC, V_{HYS} results in a noise-like precision degradation, as illustrated in Fig. 4. The corre-

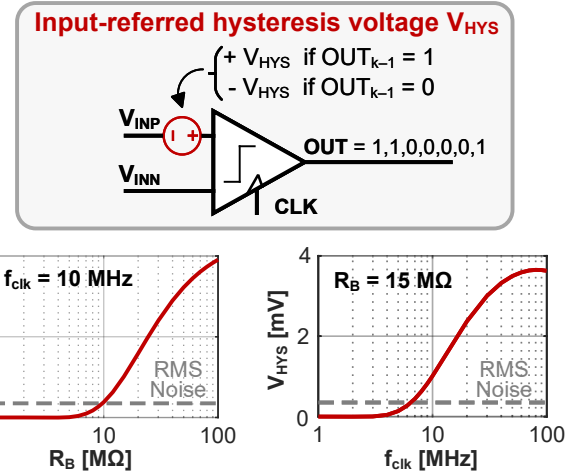


Fig. 3. Equivalent model of the hysteresis voltage V_{HYS} (top), and simulations of V_{HYS} in a conventional strongARM comparator versus bulk resistance R_B and clock frequency f_{clk} (bottom).

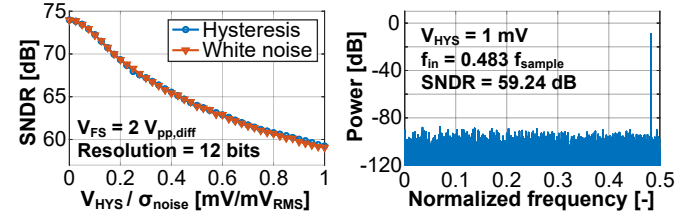


Fig. 4. Simulated performance of a 12-bit SAR ADC with a full-scale V_{FS} of $2 \text{ V}_{pp,diff}$. SNDR with hysteresis V_{HYS} or white noise σ_{noise} in the comparator (left), and output spectrum for $V_{HYS} = 1 \text{ mV}$ and $\sigma_{\text{noise}} = 0 \text{ mV}_{\text{RMS}}$ (right).

sponding hysteresis error has a white spectrum, even though it is a deterministic process, because the errors are approximately uncorrelated across consecutive ADC samples.

III. PROPOSED TRIPLE TAIL COMPARATOR WITH CAPACITIVE OVER-NEUTRALIZATION

Since the freeze-out induced hysteresis originates in the latch, a high preamplification gain can effectively suppress V_{HYS} . However, this gain is limited in the conventional SA comparator because it comprises only one preamplification stage. Prior art multi-stage comparators usually include half-latches in the preamplifier to increase the gain [13], [14]. However, these extra half-latches also suffer from the freeze-out induced hysteresis, nullifying the benefit of the additional preamplification gain.

Therefore, this work introduces the triple tail comparator with capacitive over-neutralization (TTCO) in Fig. 5. Its preamplifier consists of two stages, and the second stage (M_2) adopts the cross-coupled capacitors C_N . These capacitors boost the preamplification gain by overcompensating the gate-drain capacitance of M_2 , as is typically done in RF and mm-wave circuits [15], [16]. Cross-coupling these capacitors creates a positive feedback path for differential signals from the preamplifier outputs $V_{XP/N}$ to $V_{DP/N}$. A larger C_N provides more positive feedback, leading to a noticeably higher gain, as illustrated in the bottom-right graph of Fig. 5. In this way, the preamplification gain is significantly larger for the proposed TTCO comparator than for the conventional SA comparator.

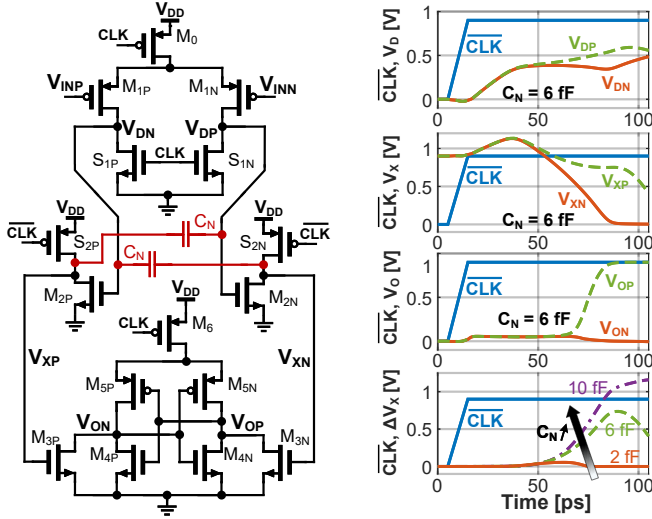


Fig. 5. Triple tail comparator with capacitive over-neutralization (TTCO) and transient simulations for $\Delta V_{IN} = 0.5$ mV, $V_{DD} = 0.9$ V, and $V_{CM} = 0.3$ V. The upper three graphs show simulations for $C_N = 6$ fF. The bottom graph shows the preamplifier output ΔV_X for $C_N = 2$ fF, 6 fF, 10 fF.

In contrast to the half-latches in prior art multi-stage comparators, the positive feedback of the cross-coupled capacitors does not cause full regeneration of the preamplifier outputs $V_{XP/N}$ to V_{DD} and V_{SS} . Instead, both V_{XP} and V_{XN} are precharged to V_{DD} during the reset phase and discharged to V_{SS} during the comparison phase. Hence, the voltage swing at the drains of M_2 causes primarily a common-mode disturbance of the bulk voltages, introducing negligible hysteresis. Consequently, the TTCO comparator is considerably less susceptible to freeze-out induced hysteresis than multi-stage comparators with half-latches in their preamplifiers.

IV. MEASUREMENT SETUP

To demonstrate the effectiveness of the proposed capacitive over-neutralization, a chip containing an SA and three TTCO comparators (C_N of 2 fF, 6 fF and 10 fF) was manufactured in a 40 nm bulk CMOS technology. All comparators were sized for similar noise and offset. To mimic the cryogenic I-V characteristics in simulations, the BSIM4 mobility and V_T model parameters were adapted based on [6], so that foundry statistical data can be reused for approximate Monte Carlo simulations [7]. To compensate for the higher V_T at cryogenic temperatures [6], [7], [8], all transistors are implemented as low- V_T (LVT) core devices. Furthermore, all comparators employ a PMOS input pair to counteract the higher impact of $1/f$ -noise at cryogenic temperatures [17]. To reduce R_B as much as possible, a double row of bulk contacts is placed as close to the latch as minimally allowed by the foundry design rules, as shown in the example layouts of Fig. 6.

The measurement setup and chip micrograph are depicted in Fig. 7. The chip is mounted on a PCB and installed in a Montana Instruments s200 cryostation for cryogenic characterization. The samples are thermalized by fixing them directly on a gold-plated copper heat sink with silver paint [7]. The minimal achievable platform temperature of this setup is 6 K. The input signals are low-pass filtered to remove the

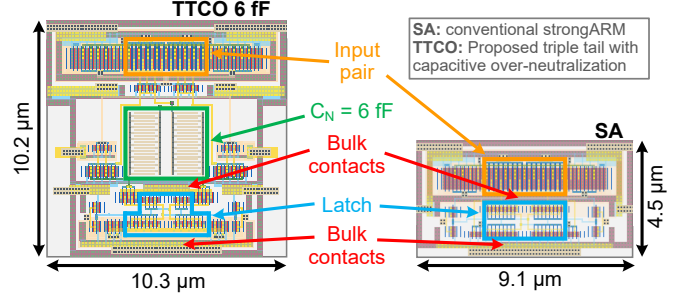


Fig. 6. Layout of the TTCO comparator with $C_N = 6$ fF (left) and the SA comparator (right).

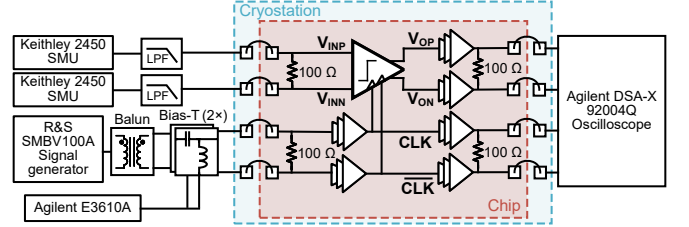
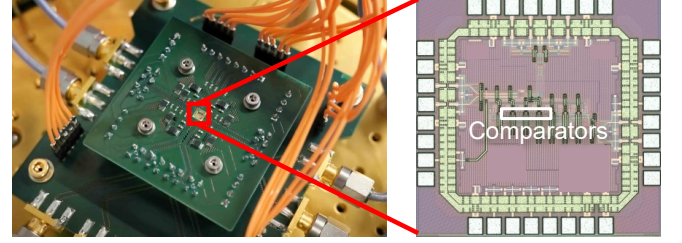


Fig. 7. Measurement PCB mounted on the base platform of a Montana Instruments s200 cryostation, chip micrograph and measurement setup.

noise of the source measure units (SMU) and the comparator and clock outputs are sampled by an oscilloscope. Unless stated otherwise, measurements are performed at a platform temperature of 6 K, a supply voltage V_{DD} of 0.9 V and an input common-mode voltage V_{CM} of 0.3 V.

V. CHARACTERIZATION METHODOLOGY

Characterizing the magnitude of the hysteresis in the presence of noise requires a statistical method, as shown in Fig. 8. The output probability of the k^{th} decision $P(\text{OUT}_k = 1)$ has a Gaussian cumulative density function (CDF) with standard deviation σ_{noise} . Without hysteresis, this CDF is centered around a differential input of 0 V (with the offset V_{OS} removed), irrespective of the previous decision. However, hysteresis shifts the CDF center to V_{HYS} or $-V_{HYS}$ when the previous decision is negative or positive, respectively, increasing the probability P_{SAME} of the same decision for the next comparison. This probability can thus be expressed using the shifted Gaussian CDF:

$$\begin{aligned}
 P_{\text{SAME}} &\triangleq P(\text{OUT}_k = 0 | \text{OUT}_{k-1} = 0) \\
 &= P(\text{OUT}_k = 1 | \text{OUT}_{k-1} = 1) \\
 &= \text{CDF}|_{\text{OUT}_{k-1}=1}(0) \\
 &= \frac{1}{2} \left[1 + \text{erf} \left(\frac{V_{HYS}}{\sqrt{2} \sigma_{\text{noise}}} \right) \right]
 \end{aligned} \tag{1}$$

TABLE I
COMPARISON OF THE PRESENTED TOPOLOGIES AT AND 6 K WITH THE STATE-OF-THE-ART HIGH-SPEED MULTI-STAGE COMPARATORS

	Cryogenic temperature (T = 6 K)				Room temperature (T = 300 K)		
	SA strongARM	This Work: TTCO			ISSCC 07 [19] Double Tail	TCAS-I 22 [13] Triple Tail	ESSCIRC 23 [14] Quadruple Tail
Technology [nm]	40	40	40	40	90	28	28
Supply voltage [V]	0.9	0.9	0.9	0.9	1.2	1.0	1.0
C _N [fF]	-	2	6	10	-	-	-
Noise voltage (σ_{noise}) [μV_{RMS}]	348	293	277	237	1500	890	980
Hysteresis voltage (V_{HYS}) [μV]	>898	>655	256	141	-	-	-
Energy/comp (E_{comp}) [fJ]	26.5	93.2	128	158	113	163	193
Delay ($V_{\text{IN}} - V_{\text{OS}} = 5 \text{ mV}$) [ps]	95	47	54	58	100 (simulated)	27	15 (estimated)
$E_{\text{comp}} \cdot \text{Delay}$ ($V_{\text{IN}} - V_{\text{OS}} = 5 \text{ mV}$) [fJ·ns]	2.5	4.4	7.0	9.2	11 (simulated)	4.4	2.8 (estimated)
Area [μm^2]	41.0	87.8	105	121	82.5	78	136
FoM _{noise} ($E_{\text{comp}} \cdot \sigma_{\text{noise}}^2$) [nJ · μV^2]	3.2	8.0	9.8	8.9	170	129	193
FoM _{HYS} ($E_{\text{comp}} \cdot V_{\text{HYS}}^2$) [nJ · μV^2]	>21.4	>40.0	8.4	3.14	-	-	-

A $P_{\text{SAME}} > 1/2$ introduces a positive correlation between consecutive comparator decisions. With the comparator biased in its trip point ($\Delta V_{\text{IN}} - V_{\text{OS}} = 0$), the mean outcome μ_{OUT} is equal to $1/2$. Then, the output autocorrelation ρ_1 across K_{TOT} decisions, defined in the following equation [18], is a measure for the increase of P_{SAME} :

$$\rho_1 \triangleq \frac{\sum_{k=1}^{K_{\text{TOT}}} (\text{OUT}_{k+1} - \mu_{\text{out}})(\text{OUT}_k - \mu_{\text{out}})}{\sum_{k=1}^{K_{\text{TOT}}} (\text{OUT}_k - \mu_{\text{out}})^2} \quad (2)$$

In the trip point, $(\text{OUT}_{k+1} - \mu_{\text{out}})(\text{OUT}_k - \mu_{\text{out}})$ equals $1/4$ with a probability P_{SAME} , while it equals $-1/4$ with a probability $1 - P_{\text{SAME}}$. Furthermore, $(\text{OUT}_k - \mu_{\text{out}})^2$ always equals $1/4$. So, for sufficiently large K_{TOT} , ρ_1 converges to:

$$\lim_{K_{\text{TOT}} \rightarrow \infty} (\rho_1) = \frac{\frac{1}{4}P_{\text{SAME}} - \frac{1}{4}(1 - P_{\text{SAME}})}{\frac{1}{4}} \quad (3)$$

$$= 2P_{\text{SAME}} - 1$$

Then, substituting (1) in (3) results in an expression for V_{HYS} :

$$V_{\text{HYS}} = \sqrt{2}\sigma_{\text{noise}} \text{erf}^{-1}(\rho_1) \quad (4)$$

VI. MEASUREMENT RESULTS

For the noise measurements, the clock is set to a low frequency of 100 kHz, resulting in the output distributions shown in Fig. 9. The σ_{noise} derived from these CDFs serves as an input for the hysteresis characterization in Fig. 10. The V_{HYS} of all four comparators is characterized for three samples and clock frequencies from 100 kHz to 50 MHz. At 100 kHz, V_{HYS} is negligible. However, around 5 MHz, the SA comparator has a V_{HYS} above 850 μV , which is almost $3\sigma_{\text{noise}}$. At higher frequencies, V_{HYS} cannot be characterized accurately anymore due to the high sensitivity of the inverse error function when the autocorrelation gets too close to 1. This means that V_{HYS} is even higher at these high frequencies. In contrast, the TTCO comparators show a noticeably lower V_{HYS} . The extra stages and the capacitive over-neutralization provide a significantly higher preamplification gain, which successfully suppresses V_{HYS} .

The temperature dependence of V_{HYS} at a clock frequency of 10 MHz is shown in Fig. 11 for the SA and the TTCO

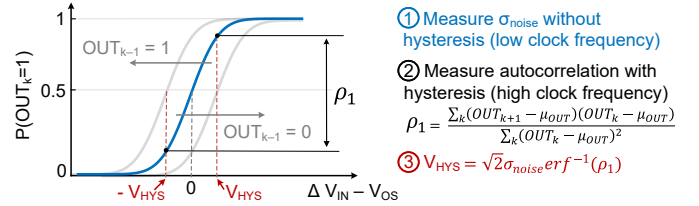


Fig. 8. Procedure to calculate V_{HYS} from the measured autocorrelation between consecutive comparator decisions.

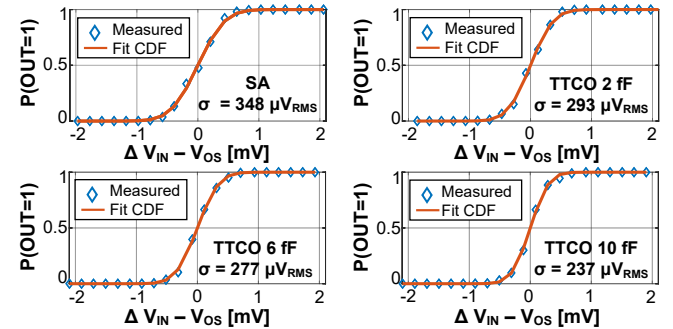


Fig. 9. Output probability distributions of the measured comparators.

comparator with a C_N of 10 fF. Below 20 K, the dopant freeze-out makes V_{HYS} rise steeply above the noise level in the SA comparator. In contrast, in the TTCO comparator V_{HYS} stays well below the noise level for all platform temperatures, down to 6 K.

Table I summarizes the measurement results at a platform temperature of 6 K. The SA comparator has a V_{HYS} above 898 μV , substantially deteriorating its precision compared to its noise level of 348 μV_{RMS} . In contrast, the TTCO comparator with a C_N of 10 fF lowers $V_{\text{HYS}} > 6\times$ to only 141 μV , well below its noise level of 237 μV_{RMS} . Since V_{HYS} degrades the SNDR of a SAR ADC like a white noise source with power $(V_{\text{HYS}})^2$, the FoM_{HYS} adapts the FoM_{noise}, introduced in [20], accordingly. Despite its higher power consumption, the TTCO comparator with a C_N of 10 fF shows a $>6\times$ better FoM_{HYS} than the SA comparator. Furthermore, the value of C_N allows to trade off V_{HYS} suppression and power consumption effectively. For completeness, the table also includes the room temperature double tail comparator from [19] and the multi-stage comparators

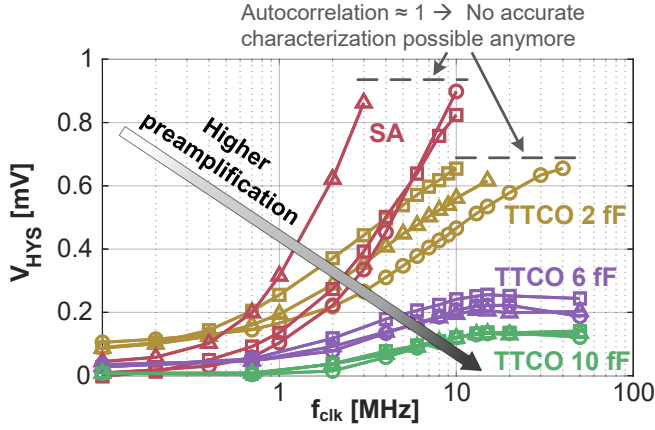


Fig. 10. Measured V_{HYS} versus clock frequency for three samples at a platform temperature of 6 K ($\square$: sample 1, $\triangle$: sample 2, $\circ$: sample 3).

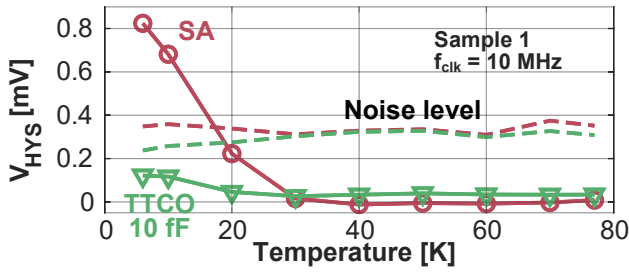


Fig. 11. Measured V_{HYS} versus platform temperature at $f_{clk} = 10$ MHz.

from [13], [14]. Even though these comparators are designed to operate under completely different conditions in terms of temperature and noise, this comparison shows that the TTCO comparators have a competitive absolute energy consumption E_{COMP} compared to state-of-the-art multi-stage comparators. Although white noise does not reduce proportional to temperature down to 6 K [9], [17], the cryogenic temperature still significantly improves the noise efficiency FoM_{noise} .

VII. CONCLUSION

The freeze-out induced increase of the bulk resistance jeopardizes a proper reset of the latch in cryogenic bulk CMOS comparators. This introduces hysteresis, favoring the same decision as the previous one. This work developed a statistical characterization procedure to reliably measure the hysteresis in the presence of noise, showing that hysteresis severely degrades the precision of the popular strongARM comparator. In contrast, the proposed TTCO comparator effectively suppresses the hysteresis in a simple and robust way by boosting the preamplification gain using capacitive over-neutralization. The prototype TTCO comparator with a C_N of 10 fF has a hysteresis voltage of only 141 μV , which stays well below its noise voltage of 237 μV_{RMS} . Hence, the TTCO comparator maintains high precision in the cryogenic environment up to clock frequencies well above 10 MHz.

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